Practical - 2

Practical Title: Semantic HTML5 Pages with Accessibility-Ready Structure

Problem Definition: Develop static HTML5 skeletons for at least 10 StudentHub pages such as Home, About, Register, Login, Dashboard, Events, Profile, Contact, Admin, FAQ, and Feedback. Use semantic tags and accessibility-friendly structure.

Theory

1. Are semantic tags such as header, nav, main, section, article, aside, and footer used properly?

Yes. Semantic HTML5 elements are used to create a clear and meaningful structure for the StudentHub pages.

- <header> is used for the page header and branding.
- <nav> is used for navigation links.
- <main> contains the primary content of each page.
- <section> is used to divide content into meaningful sections.
- <article> is used for independent content such as events, announcements, or FAQs.
- <aside> is used for additional or related information where required.
- <footer> contains footer information and useful links.

Using semantic elements improves the readability, maintainability, and accessibility of the website.

2. Are form labels, headings, image alt text, and navigation links accessible?

Yes. Accessibility-friendly HTML practices are followed throughout the StudentHub pages.

- Form inputs are associated with descriptive <label> elements.
- Headings such as <h1>, <h2>, and <h3> are arranged in a logical hierarchy.
- Images use descriptive alt attributes.
- Navigation links contain meaningful text that clearly describes their destination.

- Buttons and form controls are given clear labels.
- The page structure is designed to be understandable for users and assistive technologies.

3. Is the page structure consistent across all pages?

Yes. A consistent structure is maintained across the StudentHub pages.

The common structure includes:

<header> → <nav> → <main> → Page Content → <footer>

The navigation, branding, typography, and general layout are kept consistent so that users can easily move between different pages.

A. Implementation/ Output/Result

The StudentHub portal was implemented using semantic HTML5 elements to create accessible and well-structured static pages.

The following pages were developed:

1. Home
2. About
3. Register
4. Login
5. Dashboard
6. Events
7. Profile
8. Contact
9. Admin
10. FAQ
11. Feedback

Result

The required StudentHub pages were successfully created using semantic HTML5 elements and an accessibility-ready structure. The pages provide a consistent navigation and content structure across the portal.

B. Evidence of Work (Screenshots)

Screenshots



Student Hub Portal

ASSIGNMENT ATTENDANCE CONTACT COURSE LOGIN REGISTER RESULT DASHBOARD TIMETABLE PROFILE

Student Registration Form

Full Name:

Email:

Password:

Gender:
○ Male ○ Female

Hobbies:
☐ Reading ☐ Sports ☐ Music

City:
Ahmedabad

Address:

Submit Reset



Student Hub Portal

ASSIGNMENT ATTENDANCE CONTACT COURSE LOGIN REGISTER RESULT DASHBOARD TIMETABLE PROFILE

Student Registration

Full Name Enter Full Name Student ID Enter Student ID Email Enter Email Password Enter Password Register

Already have an account? Login



Student Hub Portal

ASSIGNMENT ATTENDANCE CONTACT COURSE LOGIN REGISTER RESULT DASHBOARD TIMETABLE PROFILE

Student Profile

Student ID	25DCE012
Name	Mukund Borole
Department	Computer Engineering
Semester	3
Email	mukund@gmail.com
Phone	9898989898



Student Hub Portal

ASSIGNMENT ATTENDANCE CONTACT COURSE LOGIN REGISTER RESULT DASHBOARD TIMETABLE PROFILE

Assignments

Subject	Assignment	Due Date	Status
Java Programming	Library Management System	20 July 2026	Pending
Data Structures	Linked List Program	22 July 2026	Completed
Computer Networks	OSI Model Report	25 July 2026	Pending
Discrete Mathematics	Graph Theory Worksheet	28 July 2026	Completed



Student Hub Portal

ASSIGNMENT ATTENDANCE CONTACT COURSE LOGIN REGISTER RESULT DASHBOARD TIMETABLE PROFILE

Weekly Timetable

Day	9-10 AM	10-11 AM	11-12 PM	1-2 PM	2-3 PM
Monday	DSA	Java	CN	Maths	Lab
Tuesday	Java	DSA	DE	CN	Lab
Wednesday	Maths	CN	Java	DSA	Lab
Thursday	DE	Maths	Java	CN	Lab
Friday	DSA	Java	Maths	DE	Sports



Student Hub Portal

ASSIGNMENT ATTENDANCE CONTACT COURSE LOGIN REGISTER RESULT DASHBOARD TIMETABLE PROFILE

Contact Us

Name Enter Your Name Email Enter Your Email Message Enter Your Message Send



Student Hub Portal

ASSIGNMENT ATTENDANCE CONTACT COURSE LOGIN REGISTER RESULT DASHBOARD TIMETABLE PROFILE

Attendance Record

Subject	Classes Attended	Total Classes	Attendance
Data Structures	27	30	90%
Java Programming	28	30	93%
Computer Networks	26	30	87%
Digital Electronics	29	30	97%
Discrete Mathematics	25	30	83%

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Student Hub Portal

ASSIGNMENT ATTENDANCE CONTACT COURSE LOGIN REGISTER RESULT DASHBOARD TIMETABLE PROFILE

Semester Result

Subject	Grade	Marks
Data Structures	A	87
Java Programming	A+	92
Computer Networks	B+	81
Digital Electronics	A	85
Discrete Mathematics	B	78

Overall CGPA : 8.50



Student Hub Portal

ASSIGNMENT ATTENDANCE CONTACT COURSE LOGIN REGISTER RESULT DASHBOARD TIMETABLE PROFILE

Enrolled Courses

Course Code	Course Name	Faculty
CE301	Data Structures	Prof. Shah
CE302	Java Programming	Prof. Patel
CE303	Computer Networks	Prof. Mehta
CE304	Digital Electronics	Prof. Desai
CE305	Discrete Mathematics	Prof. Trivedi

Description:
The screenshot shows the StudentHub authentication page containing properly labelled form fields, headings, buttons, and navigation elements. The form structure follows accessibility-friendly HTML practices.

C. GitHub Repository Link:

GitHub Repository: https://github.com/rehansaiyed72/student-hub-portal

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D. Challenges Faced:

During the implementation of the StudentHub pages, the following challenges were faced:

1. Understanding the correct use of semantic HTML5 tags instead of using elements for everything.
2. Maintaining a consistent structure and navigation across multiple pages.
3. Creating proper form labels and accessible input fields.
4. Maintaining a logical heading hierarchy on different pages.

5. Adding meaningful alternative text to images.
6. Managing common navigation and footer elements across all pages.
7. Ensuring that the HTML structure remains clean and easy to maintain.

These challenges were solved by referring to HTML5 semantic elements, following accessibility practices, and maintaining a common page structure across the StudentHub portal.

E. Learning Outcomes:

After completing this practical, I learned:

1. The purpose and proper usage of semantic HTML5 elements.
2. How to structure webpages using <header>, <nav>, <main>, <section>, <article>, <aside>, and <footer>.
3. How to create accessibility-friendly forms using labels and appropriate input elements.
4. The importance of a logical heading hierarchy.
5. How to use descriptive alt text for images.
6. How to create consistent navigation across multiple webpages.
7. How semantic HTML improves website structure, readability, and accessibility.
8. How to organize multiple static HTML pages for a larger web project.

Faculty Remarks: